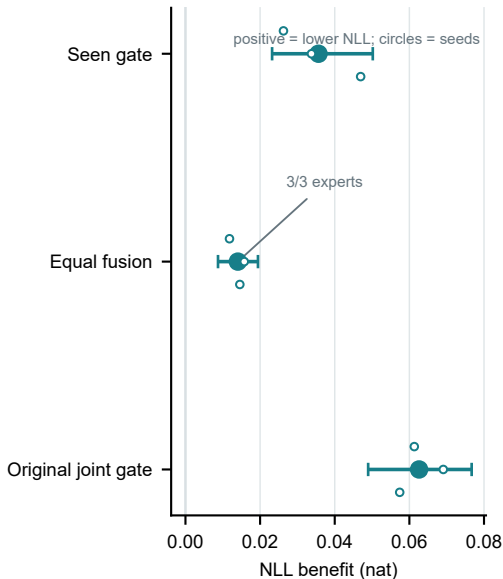
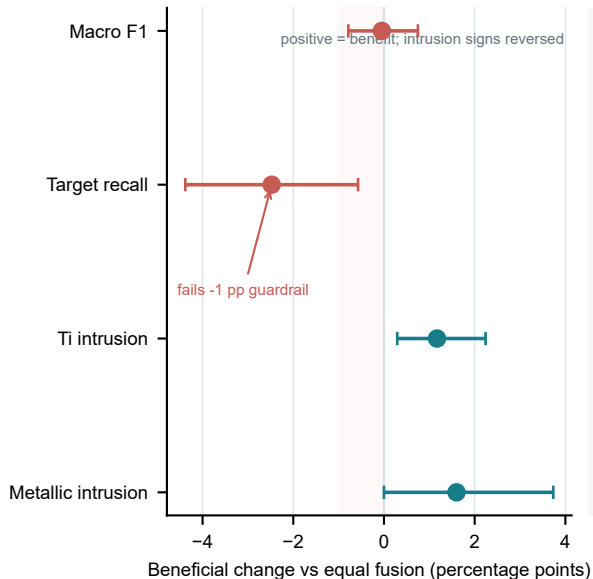


Independent OOS routing confirmation

a Proper-scoring risk



b Classification trade-off



c Preregistered decision

- PASS** NLL CI below zero
- PASS** 2/3 seed direction
- PASS** Macro F1 guardrail
- FAIL** Target recall guardrail
- PASS** Ti intrusion guardrail
- PASS** Metal intrusion guardrail

Not promoted

NLL and hard-negative control improved, but target protection was not retained.